INCH-POUND

MIL-B-49430/5G(ER)
17 November 1993
SUPERSEDING
MIL-B-49430/5F(ER)
1 August 1991

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5847/U

This specification sheet is approved for use within Army Research Laboratory , Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

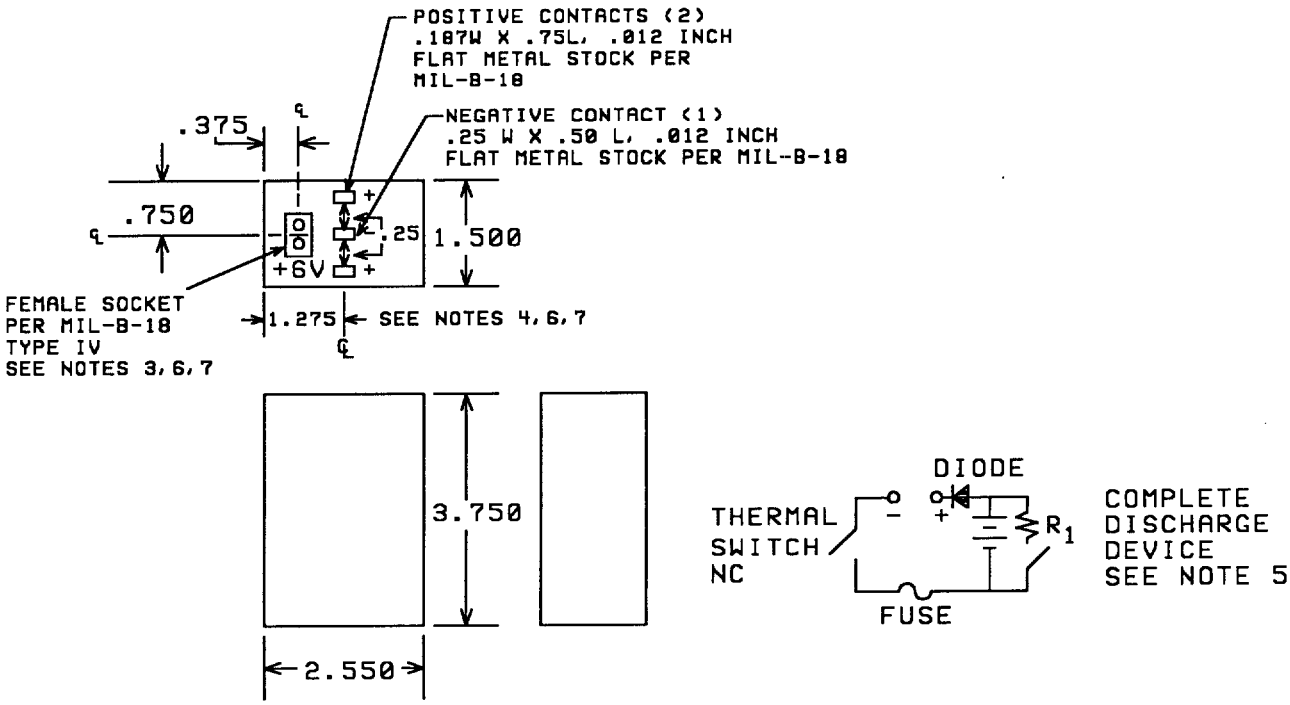


Figure 1. Battery dimensions and wiring schematic.

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NOTES:

1. All dimensions are in inches. The tolerance shall be ±0.031 inch unless otherwise specified.

2. The dual connectors shall be wired in a parallel 6-volt circuit.

3. Type IV socket:

   a) Pin circle center shall be on the battery center line.
   b) Minimum depth of the socket well shall be 5/8 inch.
   c) Shall float 1/32 inch in all directions from the pin circle center.

4. Flat contacts:

   a) The center negative contact shall be on the battery center line.
   b) Shall be supported and mounted so that the top surface shall not protrude above the outside surface of the battery jacket.
   c) Shall not be more than 1/8 inch below the outside surface of the battery jacket initially, during, and after contact is made with mating contact springs.
   d) Shall not have a lateral displacement of more than 1/32 inch during mating with contact springs.

5. R_1 value shall be between 60 ohms and 84 ohms and have a power rating of 1.0 watt.

6. Socket wiring schematic shall be printed or embossed on the battery jacket near the terminal on the terminal face of the battery.

7. Terminal markings shall be printed or embossed on the battery jacket near the connectors on the terminal face of the battery. Polarity shall be shown for each terminal as shown.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 6.10 volts.

Minimum voltage for capacity: 4.00 volts.

The battery shall contain a 2.25 ampere non-replaceable, slow blow fuse wired in series with the cells in the negative leg.

The battery shall contain a diode wired in series to prevent charging. See paragraph 3.5.5.2 of the basic specification.

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Dual Terminals: Two (2) hole female socket per MIL-B-18, Type IV; three flat contacts as per Figure 1.

Weight (maximum): 0.57 pound (255 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 4.00 Volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
T	2.8
D	2.8

Initial voltage delay: During the I, H, HT, T, and D discharge tests initial voltages below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltage below 3.80 volts cannot exceed a 60.0 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

1. Storage: Details in storage conditions for all capacity tests are specified in the basic specification (see 4.8.9).

2. Discharge:

   Cells: Discharge at 2.0 amperes constant current at 70 ± 5°F for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 5.4 hours into voltage reversal.

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Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage test: Load resistance of 5 ohms shall be used. Voltage shall be above 4.00 volts within 5 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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